

Design and Development of an Agentic System for Context-Aware Daily Mobility Services

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Abstract— Harnessing advances in agentic artificial intelligence, real-time mobility services, and vehicle automation creates new opportunities to coordinate household travel as a continuous daily mobility service rather than a series of isolated trips. This demonstration presents an agentic mobility assistant that combines bounded large language model (LLM) reasoning with deterministic calendar processing, traffic-aware routing, construction verification, electric-vehicle charging coordination, personalized output generation, and persistent route memory. The system constructs a continuous household vehicle mission from calendar and vehicle context, verifies operational details using external mobility services, and stores the resulting state for later follow-up interaction. The demonstration shows two representative household scenarios. Run 1 is a feasible six-trip day with an 80% battery, sequential caregiver drop-offs, a caregiver pickup, two protected school pickups, and no unnecessary charging or schedule intervention. Run 2 reproduces the constrained Montréal evaluation case, where low battery, a conflicting medical appointment, and construction require charging, schedule adaptation, and rerouting. The result is a persistent mobility service that can plan, verify, remember, and explain household transportation decisions.

Introduction

Modern households face growing mobility-coordination challenges—more complex daily schedules, increasingly time-sensitive caregiving responsibilities, and greater pressure to efficiently share limited transportation resources. A typical weekday may involve overlapping work commitments, medical appointments, school pickups, caregiving responsibilities, and vehicle-related tasks. When the household relies on one shared vehicle, a useful assistant must preserve high-priority responsibilities while maintaining vehicle continuity, passenger continuity, feasible departure times, and sufficient energy for the complete day.

Prior personalized travel-planning systems incorporate user schedules and preferences, while situation-aware routing shows that route choice depends on traveler and environmental context (Chiang and Huang 2015; Campigotto et al. 2017). Recent agentic systems extend travel assistance by coordinating LLM reasoning with external traffic, charging, and planning tools (Khamis 2025; Chen et al. 2026). However, these approaches primarily focus on individual users or itineraries rather than a persistent household mission that coordinates multiple calendars, caregiver assignments, protected dependents, charging, construction, user-specific visibility, and vehicle-facing operations through one shared vehicle. The present demonstration focuses on this integrated service behavior. The contribution of this demonstration is the integration of persistent household mission planning, deterministic operational verification, separate user- and vehicle-facing mission views, and route-memory interaction within one shared-mobility service.

System Overview

The prototype separates contextual reasoning from externally verifiable mobility processing. Bounded LLM agents classify requests, reason about household responsibilities and scheduling conflicts, and synthesize personalized responses. Deterministic components normalize calendars, validate and geocode destinations, generate traffic-aware route legs, verify construction impacts, evaluate charging opportunities, apply active-user filtering, and store the mission in persistent route memory. This separation prevents the LLM from independently inventing coordinates, travel times, route geometry, calendar records, or vehicle-state facts.

Execution may begin from either a user request or scheduled daily pre-computation. A planning execution constructs the household mission, verifies routing, timing, charging, and

environmental details, and produces personalized user-facing and complete vehicle-facing views. The verified mission is indexed by user and local planning date so follow-up questions can retrieve the stored state without rerunning the full pipeline unless an update is required.

Problem formulation.

For planning date t , let the household be divided into caregivers and dependents, with calendar events represented by the set E_t . Each calendar event stores the household member, time window, validated location, priority, arrival buffer, and protected-event status.

The ordered household mission and each mobility action are represented as:

$$M = \langle m_1, \dots, m_K \rangle. \quad m_k = (o_k, d_k, t_k^{dep}, t_k^{arr}, a_k, P_k),$$

Feasibility preserves vehicle continuity, temporal ordering, buffered arrivals, and the critical battery threshold:

$$d_k = o_{k+1}, \quad t_{k+1}^{dep} \geq t_k^{arr}. \\ t_k^{arr} \leq s_i - \beta_i, \quad B_k \geq B_{crit}, \quad \forall k = 1, \dots, K.$$

Candidate missions are evaluated using the following lexicographic objective:

$$\Phi(M) = (V_{protected}, V_{feasibility}, V_{conflict}, T_{travel}), \\ M^* = \underset{M \in \mathcal{M}}{\text{lex argmin}} \Phi(M).$$

Protected-event violations are minimized first, followed by caregiver, passenger, vehicle, location, and energy violations, unresolved conflicts, and finally traffic-aware travel time. The demonstration therefore tests whether the service can construct a feasible shared-vehicle mission while preserving protected dependent responsibilities.

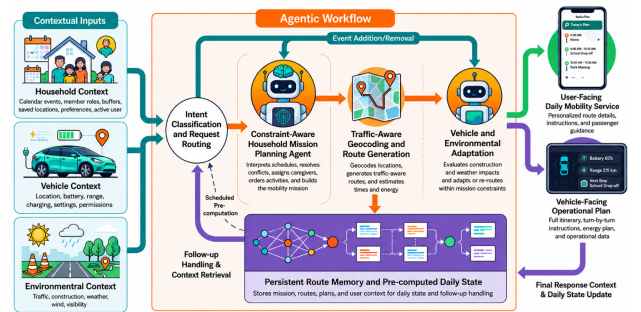


Figure 1: Agentic mobility service architecture. Household, vehicle, and environmental context are coordinated through bounded LLM reasoning, deterministic mobility processing, and persistent route memory to produce user- and vehicle-facing outputs.

Demonstration

The live demonstration uses a conversational interface connected to the workflow through webhooks. Google Calendar provides household schedules; TomTom supports geocoding and traffic-aware routing; Montréal construction records support roadwork verification; Circuit électrique information supports charging decisions; and weather is retained as context. Before Run 1, the interface exposes the household, vehicle, buffer, routing, and weather settings that shape the generated mission.

Source code, setup instructions, and implementation materials are available at:
<https://github.com/olibreadstick/agent-mobility-assistant>

Representative Run 1: feasible shared-vehicle day.

Run 1 demonstrates a feasible day with two caregivers, two dependents, one shared electric vehicle, and Parent1 as the active user. Parent2 has a 9:00 AM work meeting, Parent1 a 10:00 AM research meeting, and Child1 and Child2 protected school pickups at 3:30 PM and 4:15 PM. The vehicle begins at 80% battery, above the preferred 65% threshold, so no charging intervention is required.

The user-facing plan contains six driving trips: Home → Parent2 workplace → Parent1 workplace → Parent2 workplace for caregiver pickup → Child1 school → Child2 school → Home. The first two legs preserve shared-ride continuity, and Parent1's same-location pickup after the research meeting is represented as an on-site transition rather than a zero-distance driving leg. The run demonstrates caregiver coordination, protected dependent responsibilities, passenger continuity, traffic-aware timing, and battery-aware planning without unnecessary intervention.

Representative Run 2: constrained Montréal evaluation case.

Run 2 reproduces the constrained Montréal evaluation case. The vehicle begins at 35% battery, below the preferred 65% threshold, while Parent1 must preserve two protected school pickups despite a conflicting 4:00 PM medical appointment. The workflow inserts autonomous charging during an available idle period and recommends rescheduling the medical event rather than moving the protected pickups. The complete vehicle-facing plan contains eight legs covering approximately 17.9 km and 1 h 19 min. A Rue Mansfield construction overlap is verified against route geometry, and an alternative through McGill College Avenue is retained within the configured detour limits.

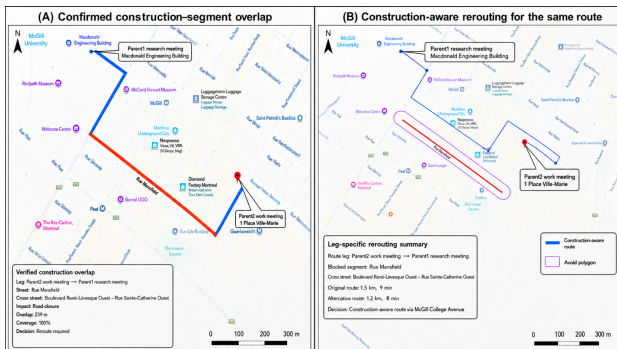


Figure 2: Construction-aware adaptation in Run 2. The preliminary route overlaps the reported Rue Mansfield work zone; the retained alternative avoids the affected segment through McGill College Avenue.

Persistent interaction.

After either mission is stored, route-memory questions can retrieve the current plan, protected-pickup departure times, or

vehicle battery level without rerunning the full planning pipeline. The same verified daily state therefore supports both the structured dashboard and concise conversational answers.

Interactive capabilities and scope.

The interface supports route planning, route follow-up, calendar-event addition and removal, and casual interaction. New route requests activate the complete mission workflow, while valid follow-ups use stored state for the same user and date. The dashboard presents weather, conflicts, trips, caregiver responsibilities, battery state, and charging information.

A complete vehicle-facing plan is preserved separately so personalization does not remove operational movements. It includes passenger-carrying and passenger-free travel, charging, caregiver and dependent pickups, and return-home operations; in this prototype it remains simulated workflow output rather than direct vehicle control.

Conclusion

The demonstration shows how a conversational trip-planning assistant can operate as a persistent daily mobility service for a shared household vehicle. Run 1 coordinates a feasible six-trip day without unnecessary intervention, whereas Run 2 shows adaptation to low battery, a protected-pickup conflict, and construction. Across both scenarios, the system grounds operational quantities in deterministic mobility services and supports separate user- and vehicle-facing views with persistent route memory.

Using GPT-5.2, the complete system achieved 94.44% mean route accuracy, compared with 75.00% for an intent-guided LLM-only baseline and 72.22% for a calendar-tool agent. The largest differences occurred in verified alternatives, active-user filtering, and schedule feasibility. Although the reduced systems were faster, they omitted traffic-aware routing, construction verification, charging coordination, user projection, and persistent route memory. These results indicate that the hybrid workflow adds operational capabilities beyond high-level calendar reasoning alone.

References

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